Conceptual Schema - E-Commerce Application

1. Introduction

This document shows the conceptual schema of our e-commerce application.

The conceptual schema is the first step of the database design. Here we only describe the main "things" the store needs to remember (entities), what information we keep about each one (attributes), and how they are connected (relationships).

We do not include primary keys, foreign keys or data types yet. Those will be added in the logical schema and in the SQL Server implementation.

2. Entities

Entity	What it represents
User	An account used to log in to the application
Role	The type of user: ADMIN, VENDOR or CUSTOMER
Vendor	A company that sells products in the store
Customer	A person who buys products
Address	A shipping address of a customer
Category	A group of products (for example Electronics or Books)
Product	An item that is sold in the store
Order	A purchase made by a customer
OrderItem	One line of an order (which product and how many)
Invoice	The bill created when an order is completed
Payment	The money paid for an order

The project asks for at least Product, Vendor, Customer, Order, OrderItem, Invoice and Payment. We added User, Role, Category and Address because we need them for the login, to organize the products and to know where to ship the orders.

3. Attributes

Attributes marked with (unique) cannot be repeated between two records. For example, two users cannot have the same email.

Entity	Attributes
User	username (unique), email (unique), password, enabled
Role	name (unique)
Vendor	company name (unique), phone, active
Customer	first name, last name, phone
Address	street, city, province, postal code, country
Category	name (unique), description
Product	SKU (unique), name, description, price, quantity in stock, status
Order	order date, status, total
OrderItem	quantity, unit price
Invoice	invoice number (unique), issue date, total
Payment	payment method, amount, payment date

Notes:

- The email is only stored in User, so it is not repeated in Vendor or Customer.
- The password will never be saved as plain text. It will be saved encrypted (hashed).
- The product status (IN_STOCK or OUT_OF_STOCK) depends on the quantity in stock.
- The order total is calculated from the order items (quantity x unit price).

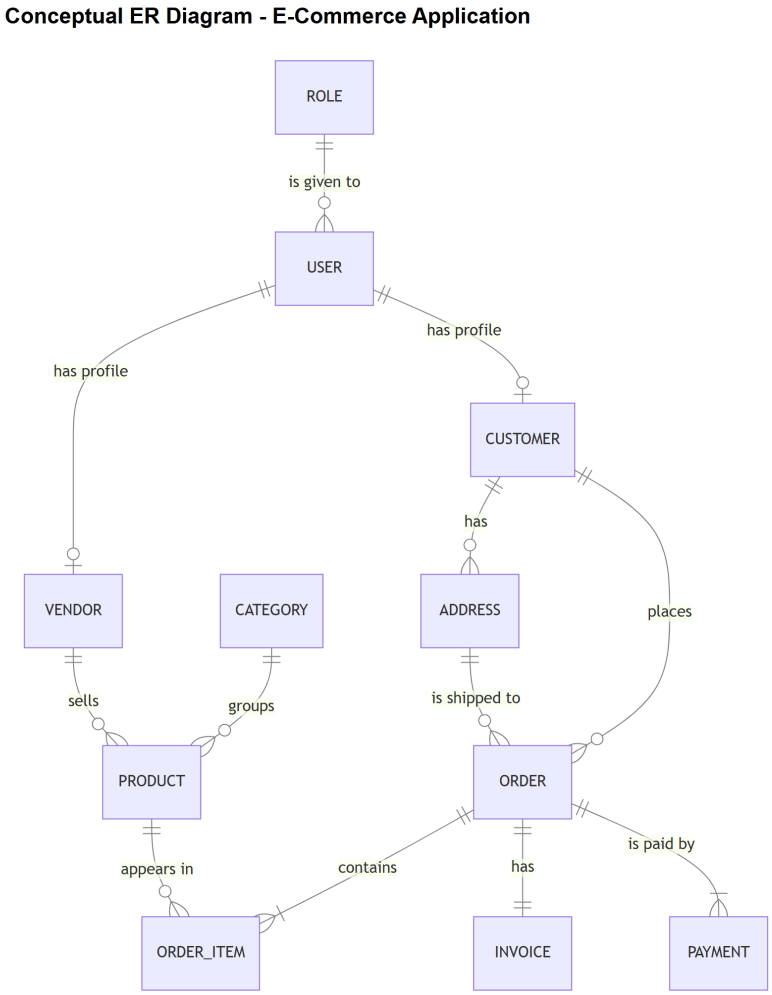
4. Relationships

Relationship	Type	Explanation
Role - User	One to many	A role can be given to many users. Each user has only one role.
User - Vendor	One to one (optional)	A vendor has one user account to log in.
User - Customer	One to one (optional)	A customer has one user account to log in. An admin is a user with no vendor or customer profile.
Vendor - Product	One to many	A vendor can sell many products. Each product belongs to one vendor.
Category - Product	One to many	A category has many products. Each product is in one category.
Customer - Address	One to many	A customer can save many addresses. Each address belongs to one customer.
Customer - Order	One to many	A customer can make many orders. Each order belongs to one customer.
Address - Order	One to many	Each order is sent to one address. The same address can be used for many orders.
Order - OrderItem	One to many	An order has one or more items. Each item belongs to one order.
Product - OrderItem	One to many	A product can appear in many order items. Each item is about one product.
Order - Invoice	One to one	Each order has one invoice and each invoice belongs to one order.
Order - Payment	One to many	An order can be paid with one or more payments. Each payment belongs to one order.

Why is the Order - Invoice relationship one to one?

In our application a purchase is saved completely or not at all. So every order that exists in the database was completed and always has its invoice.

5. ER Diagram



How to read the diagram

The diagram uses the crow's foot notation. The symbol at the end of each line tells how many records can be connected:

Symbol	Meaning
Two lines	Exactly one
Circle and line	Zero or one
Line and crow's foot	One or many
Circle and crow's foot	Zero or many

For example, between CUSTOMER and ORDER: on the customer side there are two lines, so each order belongs to exactly one customer. On the order side there is a circle and a crow's foot, so a customer can have zero or many orders (a new customer has not bought anything yet).

6. Business Rules

1. The price of a product must be greater than 0.
2. The quantity in stock cannot be negative.
3. When the quantity of a product is 0, its status is OUT_OF_STOCK. Otherwise it is IN_STOCK.
4. Usernames, emails, company names, category names, SKUs and invoice numbers cannot be repeated.
5. An order must have at least one item, and the quantity of each item must be more than 0.
6. A customer can only buy a product if there is enough stock and the vendor is active.
7. An order can only be shipped to an address of the same customer.
8. The payment must cover the total of the order. If not, the purchase is rejected.
9. A purchase is all or nothing. The order, the items, the stock update, the invoice and the payment are saved together. If one step fails, nothing is saved (rollback).
10. Two customers cannot buy the last unit of the same product at the same time.
11. The payment method can be CREDIT_CARD or BANK_TRANSFER.
12. Only an ADMIN can activate or deactivate a vendor.
13. Prices and totals use exact decimal numbers (not float or double).